ACCESS · INDIGENOUS SKILLS 4 SUCCESS

Teaching and answer guidance · Compaction and final handover

Volume 8 · L1, L2, L3, L4, L5, L6

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In this volume

1. How to use the volume and key terms
2. Compaction equipment
3. Compaction work safety procedures
4. Asphalt Compaction Roller operator roles and responsibilities
5. Compaction Roller maintenance
6. Operate Compaction Roller
7. Follow shut down procedures
8. Read the rolling record and its limits

Use this volume

This volume provides complete classroom readings, worked examples, named practice tasks and answer guidance for compaction and final handover. Follow the learning route in each chapter, study the labelled figures and explain your reasoning. The separate answer book contains the worked keys and observation records.

Care for people, place and the next person using the work runs through the cases. Local First Nations, Métis and Inuit learners and partners bring distinct knowledge and priorities. Participation does not require sharing personal stories or cultural teachings, and no diagram claims to represent a Nation's protocol.

> Classroom authoring is complete for the listed competencies. Qualified technical review, actual fleet/site matching, a local teaching pilot and observed practical qualification are separate requirements. The practical records are ready for an instructor to configure and use; they do not certify unobserved work.

Read the picture, then test the idea

1. Identify the object, process or quantity shown.
2. Read labels, units and the figure's stated limits.
3. Connect the image to the chapter explanation.
4. Try the exercise and check against evidence.
5. Explain the result to another person and revise if needed.

Use AI as a checked working tool

Claude and ChatGPT can help explain provided documents and create supported files. Claude Code, Cowork and Codex can help with permitted multi-file tasks and repeatable workflows when their tools are available. Give a clear purpose, named sources, output formats, limits and checks. Ask for a findings table with page/cell evidence; verify it against the source. Reopen DOCX, XLSX, HTML or PDF outputs. For study audio/video, verify the script, captions, diagrams and actual playback. A generated result is not a technical approval.

Keep the term with its meaning

AGGREGATE
Mineral particles forming the solid skeleton of the mixture or granular layer.

BINDER
The asphalt material that coats and binds the aggregate in the specified mixture.

COMPACTED THICKNESS

Layer thickness after the required compaction process; distinct from loose depth behind the screed.

DENSITY REFERENCE
The specified comparison value used as the denominator of a percent-density calculation.

GRADATION
Distribution of aggregate particle sizes, commonly expressed as cumulative percent passing specified sieves.

HEAD OF MATERIAL
The quantity/distribution of mix immediately in front of the screed.

JOB MIX FORMULA (JMF)
The accepted mixture proportions and properties under the applicable quality/contract process.

LOCKOUT
Isolation, securing and verification of hazardous energy for work requiring protection.

LOT / SUB-LOT
Defined portions of work used for sampling, testing and acceptance under the contract.

OMM
Operation and maintenance manual matched to the actual machine, serial range and configuration.

RAP
Reclaimed asphalt pavement: recovered material containing aggregate and aged binder.

SDS
Safety data sheet for the actual product; read with its label and workplace procedure.

SCREED
The paver assembly that forms the mat; its response depends on setup, material and movement.

STATION / OFFSET
Location along an alignment / distance and direction from a defined reference.

TCP
Traffic control person with the required training and assigned role.

TRACEABILITY
A record connecting material, location, time, sample, result and authorized decision.

SVG
Scalable vector graphics: useful for clear diagrams whose labels and geometry must be checked.

DOCX / XLSX
Editable document / spreadsheet workbook; verify actual content and formulas after creation.

PDF / HTML
Page-based reading copy / browser document; inspect completeness, access and linked resources.

MP3 / MP4 / VTT

Audio file / video container / timed web captions; test playback and retain a transcript.

Compaction equipment

Worked example · Choose by the task

A narrow access area has restricted turning space, while the main road has a high production rate.

Compare suitable machine dimensions, modes, capacity and support for each area, then coordinate the planned roles and testing. Do not select only by the largest operating mass or assume one roller can maintain both areas without a scheduling conflict.

Explain and apply the idea

Identify drum, scraper, spray bar, articulation joint, ROPS and seat belt on the supplied model figures.

Answer guidance.

Use the actual visible labels and explain: drum contacts/compacts; scraper manages buildup; spray system wets as designed; articulation steers; ROPS/seat belt form the fitted protective arrangement. Stay clear of articulation hazards.

Observe: Reading, Communication through the learner’s actual contribution, calculation, explanation or saved work.

A fictional effective coverage width is 1.4 m over a 70 m strip

A fictional effective coverage width is 1.4 m over a 70 m strip. Find area for one full-length coverage strip.

Answer guidance.

1.4 × 70 = 98 m². Actual overlap, pass definitions and rolling pattern must be specified; this is geometry, not a density result.

Observe: Numeracy, Problem solving through the learner’s actual contribution, calculation, explanation or saved work.

Ask an AI to compare static, vibratory and oscillatory action, then audit its table.

Ask an AI to compare static, vibratory and oscillatory action, then audit its table.

Answer guidance.

Require source-backed definitions and distinguish mechanism from recommended settings. Verify against the supplied guides; reject universal claims about pass counts, frequency or density acceptance.

Observe: Digital, Writing, Creativity and innovation through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Identify roller families and actual components.	0 / 1 / 2	Evidence/help: _____
Explain breakdown, intermediate and finish roles.	0 / 1 / 2	Evidence/help: _____
Distinguish monitoring displays from acceptance tests.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Compaction work safety procedures

Worked example · A shortcut behind the roller

A worker steps behind a roller to shorten the route while the operator is preparing to reverse.

Stop the affected movement and re-establish the controlled pedestrian route and communication. Review the blind-area conflict with the crew. High-visibility clothing does not authorize crossing an active reversing path.

Explain why “three metres is always safe” is an inadequate rule.

Explain why “three metres is always safe” is an inadequate rule.

Answer guidance.

Safe separation depends on machine size, movement, visibility, stopping distance, ground conditions and the site plan. Use the actual assessment and instructions; a generic number can be unsafe.

Observe: Problem solving, Reading through the learner’s actual contribution, calculation, explanation or saved work.

A route change reduces available clearance by 0.6 m

A route change reduces available clearance by 0.6 m. What should be recorded?

Answer guidance.

Original and revised geometry, machine/movement envelope, affected people/routes, assessment and authorized decision. The subtraction alone cannot establish safe clearance.

Observe: Numeracy, Writing through the learner’s actual contribution, calculation, explanation or saved work.

Demonstrate a clear stop and read-back with the paver lead.

Demonstrate a clear stop and read-back with the paver lead.

Answer guidance.

Use machine ID, location and the specific conflict; obtain acknowledgement, maintain safe positions and confirm the restored plan before resuming.

Observe: Communication, Collaboration, Adaptability through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Inspect/use fitted ROPS and belt and recognize blind/articulation zones.	0 / 1 / 2	Evidence/help: _____
Demonstrate the applicable lockout and frame-lock procedure.	0 / 1 / 2	Evidence/help: _____
Park and secure with required chocks/controls and access protection.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Asphalt Compaction Roller operator roles and responsibilities

Worked example · Pass count without coverage

Two operators each report six passes, but their paths overlap on one side and leave a strip untreated.

Reconstruct the actual path and coverage map, identify the missed strip and obtain an authorized response while conditions allow. Define pass/coverage consistently for the next run. Twelve reported movements do not establish uniform compaction.

Draw a conceptual coverage map with three labelled strips and a legend for complete/needs check.

Draw a conceptual coverage map with three labelled strips and a legend for complete/needs check.

Answer guidance.

Use patterns or labels as well as colour, identify direction and boundaries, and record the actual approved sequence separately. Do not present the sketch as a universal rolling pattern.

Observe: Writing, Creativity and innovation through the learner’s actual contribution, calculation, explanation or saved work.

A training record has four strips, each requiring two planned coverages

A training record has four strips, each requiring two planned coverages. How many strip-coverages are planned?

Answer guidance.

4 × 2 = 8 strip-coverages. This exercise count is not a recommended field pass requirement or proof of density.

Observe: Numeracy, Problem solving through the learner’s actual contribution, calculation, explanation or saved work.

Write a handover for a strip awaiting a quality check.

Write a handover for a strip awaiting a quality check.

Answer guidance.

Identify lane/station/strip, actual completed treatment, known temperature/time, pending test and person responsible. Confirm where other equipment and people may move.

Observe: Communication, Collaboration, Digital through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Coordinate multiple operators using clear location and coverage terms.	0 / 1 / 2	Evidence/help: _____
Follow and record the approved pattern.	0 / 1 / 2	Evidence/help: _____
Recognize and report ride/texture concerns without unsupported corrections.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Compaction Roller maintenance

Worked example · A dry stripe on the drum

A drum shows a persistent dry stripe and material begins to pick up in that area.

Report the pattern, stop/hold as required and inspect the spray/scraper system through the safe procedure. A blocked nozzle or another fault is a possibility, not a confirmed diagnosis. Correct and verify the authorized repair before continuing.

A fictional 120 L tank contains 35 L

A fictional 120 L tank contains 35 L. How much water reaches the stated 120 L level?

Answer guidance.

85 L, assuming those stated values and no other limits. Use the actual machine filling instructions and do not infer a recommended tank capacity.

Observe: Numeracy, Reading through the learner's actual contribution, calculation, explanation or saved work.

List four water-system items for a model-specific winterization record.

List four water-system items for a model-specific winterization record.

Answer guidance.

Tank, pump, filters/lines and spray bars/nozzles as fitted, with each actual drain/protection step and verification. Use the complete OMM procedure.

Observe: Writing, Problem solving through the learner's actual contribution, calculation, explanation or saved work.

Build a maintenance checklist with AI assistance and audit it before use.

Build a maintenance checklist with AI assistance and audit it before use.

Answer guidance.

Give the matching manual/serial scope, ask for page references, verify every item and interval, flag missing access or unread pages, and have the responsible qualified person approve the practical checklist. AI does not supply technical authorization.

Observe: Digital, Communication, Collaboration through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

The outline specifies a minimum 80% checklist result for this task. Score the 3 task criteria out of 6; require at least 5 points and all critical items passed. This local scoring rubric operationalizes the checklist; it is not a separate certification. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Service lubrication/intake/separator and correct fluids safely.	0 / 1 / 2	Evidence/help: _____
Inspect cooling/battery and fitted safety systems.	0 / 1 / 2	Evidence/help: _____
Maintain sprays, scrapers/contact surfaces and model-specific winterization.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Operate Compaction Roller

Worked example · Density feedback changes the plan

A trial strip's recorded density increases over the first approved passes, then stops improving while the mixture cools.

Report the measured results with temperature, location and reference. The quality/crew lead evaluates whether settings, timing, equipment or another factor needs adjustment. More passes are not automatically effective and may damage the mat. Preserve the trial record and authorized revised pattern.

In-place density is 2.310 Mg/m³ and the required maximum-theoretical reference is 2.500 Mg/m³

In-place density is 2.310 Mg/m³ and the required maximum-theoretical reference is 2.500 Mg/m³. Calculate percent.

Answer guidance.

2.310/2.500 × 100 = 92.4%. Interpretation requires the applicable method, lot and contract limits; it is not a standalone acceptance decision.

Observe: Numeracy, Reading through the learner's actual contribution, calculation, explanation or saved work.

An infrared reading and probe reading differ

An infrared reading and probe reading differ. What should be checked before declaring either wrong?

Answer guidance.

Instrument identity/calibration, method, location, timing, surface versus internal measurement, units and specified procedure. Record both and seek the responsible person's interpretation.

Observe: Problem solving, Communication through the learner's actual contribution, calculation, explanation or saved work.

Create a rolling record that another operator can continue from.

Create a rolling record that another operator can continue from.

Answer guidance.

Include machine/role, lane/station/strip, approved pattern/settings, actual coverages, time/temperature, test references, changes/authorization and pending areas. Distinguish observations from assumptions.

Observe: Digital, Writing, Collaboration, Adaptability through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Precheck, service, start/monitor and shut down the actual roller.	0 / 1 / 2	Evidence/help: _____
Perform flat forward/stop/reverse/stop, parking and travel-mode controls safely.	0 / 1 / 2	Evidence/help: _____
Operate fitted vibration through the approved sequence.	0 / 1 / 2	Evidence/help: _____
Demonstrate assigned confined-edge and longitudinal-joint patterns.	0 / 1 / 2	Evidence/help: _____
Coordinate breakdown/intermediate/finish roles with the paving train.	0 / 1 / 2	Evidence/help: _____
Interpret temperature/density feedback and maintain a traceable coverage record.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Follow shut down procedures

Worked example · Ready for the next shift

A roller is clean and fuelled but parked where a delivery truck must reverse around it beside a public entrance.

Have the responsible lead arrange a safer approved staging location. Complete the movement and shutdown under the actual procedure, preserve access and communicate the final location/status. “Ready” includes safe placement, not only clean equipment.

State the difference between staging and transport securement.

State the difference between staging and transport securement.

Answer guidance.

Staging positions equipment safely for site access and the next task. Transport securement restrains the configured load on a carrier under manufacturer and applicable transport rules. Both require planned parking/movement controls, but their arrangements differ.

Observe: Reading, Communication through the learner’s actual contribution, calculation, explanation or saved work.

A ten-item checklist has 20 possible points

A ten-item checklist has 20 possible points. A learner earns 18 but fails a critical roll-away control. Is the outcome satisfactory?

Answer guidance.

90% numerically, but not satisfactory for release. Correct and reassess the critical control; the 80% outline threshold does not permit an unsafe final state.

Observe: Numeracy, Problem solving through the learner’s actual contribution, calculation, explanation or saved work.

Write a final shift handover with a pending spray-system repair.

Write a final shift handover with a pending spray-system repair.

Answer guidance.

Identify roller/meter, parking location and secure state, cleaning/service completed, exact spray defect, hold/out-of-service status and responsible person/next action. Do not describe it as ready for production until authorized.

Observe: Writing, Digital, Collaboration through the learner's actual contribution, calculation, explanation or saved work.

Instructor observation record

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

The outline specifies a minimum 80% checklist result for this task. Score the 4 task criteria out of 8; require at least 7 points and all critical items passed. This local scoring rubric operationalizes the checklist; it is not a separate certification. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Perform matching shutdown, parking and required chocking.	0 / 1 / 2	Evidence/help: _____
Clean safely and stage without blocking access.	0 / 1 / 2	Evidence/help: _____
Prepare transport configuration and communicate roles.	0 / 1 / 2	Evidence/help: _____
Secure the machine and hand over defects/readiness.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Integrated practice · all nine Skills for Success

Read the rolling record and its limits

> This is a complete fictional classroom case. It does not describe an actual community, Nation, employer or project. Learners may use the supplied case without disclosing personal or cultural knowledge.

A fictional density record gives in-place 2.325 Mg/m³ and maximum theoretical 2.500 Mg/m³. A coverage sketch marks strips A and B complete and C awaiting a check. A draft says “93% means the whole lot is accepted” and “six passes always work.” No contract limit, sub-lot results or approved fixed pass count is supplied.

Reading · Check the facts

Identify the controlling facts, missing information and one unsupported statement in the brief.

Evidence and feedback: Use the exact case facts; distinguish observation, assumption and missing authority.

Writing · Make the handover useful

Draft a concise handover with location, quantities, status and the next responsible action.

Evidence and feedback: The note preserves verified facts, units and unresolved holds without inventing a release.

Numeracy · Calculate and check

Show the case calculation with units and an independent reasonableness check.

Evidence and feedback: Use the worked numerical key below; show the relationship and its practical limits.

Digital · Direct AI and verify the files

Use an available AI assistant to draft a findings table and editable record from this brief. Request DOCX or a structured document, XLSX with formulas where useful, and a PDF/HTML reading copy. Audit every claim and reopen the actual outputs.

Evidence and feedback: Record the product/capabilities, prompt, source facts, saved files and checks. A chat response alone is not a created or verified file. Check formulas and preserve originals.

Problem solving · Investigate the discrepancy

Explain the discrepancy, what evidence is needed and who should authorize the next step.

Evidence and feedback: Separate the observed issue from a guessed cause; preserve evidence and identify the correct decision role.

Communication · Send and check the message

Brief a partner, ask for a check-back and repair one misunderstanding if it occurs.

Evidence and feedback: Observe an actual exchange and confirmation of the essential facts; do not infer communication from a shared file.

Collaboration · Review together

Agree reviewer/author roles, complete a review and record each person’s contribution.

Evidence and feedback: Both learners take an identifiable turn; feedback is respectful and tied to the case evidence.

Adaptability · Respond to the change

Apply the change card, identify what changes and what stays, and revise the record.

Evidence and feedback: Use the change-response key below; retain the earlier version and do not alter unrelated facts.

Creativity and innovation · Make understanding easier

Create a labelled SVG, annotated plan, short audio explanation or captioned storyboard that makes the issue easier to understand. Test it with a reader/listener and improve one useful detail.

Evidence and feedback: The visual/media has accurate labels, source credits and a stated purpose. Inspect generated diagrams/media; do not invent technical settings. Keep a transcript/captions when producing audio/video.

Change card · Use after the first attempt

The spray system develops a dry stripe and the operator observes material pickup.

Verified case key

Percent density = 93.0% using the stated maximum-theoretical reference. It does not establish whole-lot acceptance. Preserve the pending strip and actual trial/temperature/coverage evidence; reject a universal six-pass claim.

Change-response key

Report and hold/stop as required, investigate the actual spray/scraper system safely and obtain authorized correction before resuming.

Individual evidence record

Keep the brief, calculation, saved work, review findings and revision. Record each person’s actual contribution and support. Use Not observed, Not yet demonstrated, Demonstrated with support, or Demonstrated independently for the stated action; these are local feedback descriptions.

For AI/media work, a transcript, script or storyboard is distinct from a generated audio/video file. Name what was actually produced and inspected. Do not upload private records or community-held material without the relevant authority.